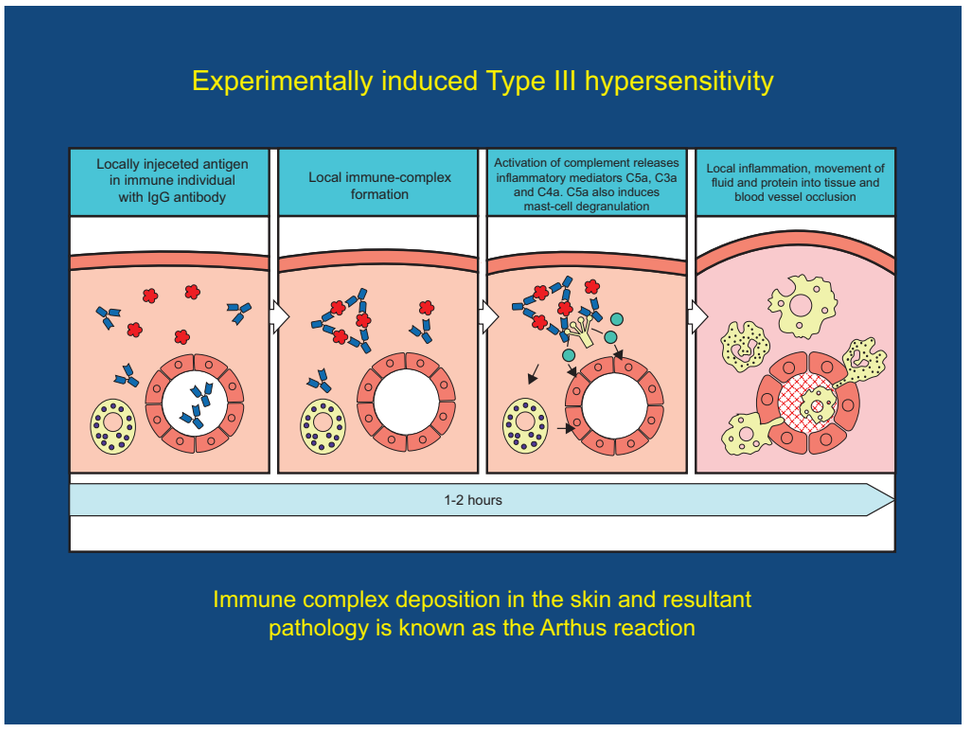


Figure 11.9: Type III hypersensitivity

necrosis. This reaction occurs when antibody is found in excess. It appears in 2–8 hours after injection and persists for about 12–24 hours (Table 11.1).

Type IV hypersensitivity: Cell Mediated Delayed Hypersensitivity

It is often called as delayed hypersensitivity reaction as the reaction takes two to three days to develop. Type IV hypersensitivity is involved in the pathogenesis of many autoimmune and infectious diseases such as tuberculosis and leprosy. T lymphocytes, monocytes and macrophages are involved in the reaction. Cytotoxic T Cells cause direct damage whereas the T helper cells secrete cytokines and activate monocytes

Table 11.1: Difference between Immediate Hypersensitivity and Delayed Hypersensitivity

Sn. No.	Immediate Hypersensitivity	Delayed Hypersensitivity
1.	It appears and disappears rapidly	It appears slowly and last longer.
2.	It is induced by antigens or haptens by any route	Induced by infection, injection of antigen intra dermally or with adjuvants of by skin contact.
3.	The reaction is antibody mediated B-cell response	The reaction is T-cell mediated response.
4.	Passive transfer is possible with serum	Cannot be transferred with serum but can be transferred by lymphocytes
5.	Desensitization is easy, but does not last long	Desensitization is difficult but long lasting.

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